

Chapter 1: Artificial Intelligence

Artificial Intelligence (AI) enables machines to mimic human intelligence through learning, reasoning, problem solving, and decision making.

Key Points

- Definition
- Real-world applications
- Advantages
- Interview tips
- Project relevance

Chapter 2: Machine Learning

Machine Learning is a subset of AI that allows computers to learn patterns from data and make predictions.

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Chapter 3: Deep Learning

Deep Learning uses neural networks with many hidden layers to solve complex tasks such as vision and NLP.

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Chapter 4: Large Language Models

LLMs like Llama, GPT, Gemini, and Mistral generate and understand human language using transformer architectures.

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Chapter 5: Generative AI

Generative AI creates text, images, code, music, and other content from user prompts.

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Chapter 6: Embeddings

Embeddings convert text into dense vectors so similar meanings are close together in vector space.

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Chapter 7: Vector Databases

Vector databases such as ChromaDB and Pinecone store embeddings and perform semantic similarity search.

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Chapter 8: Retrieval-Augmented Generation

RAG retrieves relevant document chunks from a vector database before asking the LLM to answer.

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Chapter 9: LangChain

LangChain helps developers build LLM applications using prompts, memory, chains, tools, agents, and RAG.

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Chapter 10: AI Agents and Automation

AI agents use LLMs, APIs, memory, and tools to perform multi-step tasks automatically.

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